

Yeshwantrao Chavan College of Engineering, Nagpur

Project Preliminary Investigation Report

Name of Department:

Computer Science and Engineering

Name of Project Guide:

Prof. Chanchla Tripathi

Students Details :

Roll No.	Name of Student	Email ID	Mobile No.
01	Hemanshu Waghmare	hemanshu.waghmare@gmail.com	9284133409
02	Rishabh Jain	rjain23112002@gmail.com	9322192806
03	Dhruv Dalvi	dhruvdalvi786@gmail.com	7378749332
04	Sanket Asole	sanketasole786@gmail.com	7796004860
05	Yuvraj Chavan	yuvichavan968@gmail.com	9309119399

Title of the Project :

Speak2Summarize

Area of Project Work :

Natural Language Processing (NLP) and Machine Learning for Productivity Enhancement

Problem Statement :

Many people find it challenging to remember daily tasks and important details, leading to decreased productivity, missed responsibilities, and heightened stress. An effective solution is crucial for improving task management and memory recall.

Prior Art (Patent Search) :

Patent Application No.	Title of Patent	Existing Solutions (Abstract of Patent)
18/540,594	MULTI-MODAL VOICE RECOGNITION SYSTEM AND METHOD FOR CONVERSATION SUMMARIZATION	Disclosed herein is a voice recognition system with an enhanced summarization function according to the present invention. The voice recognition system include: an audio feature extractor configured to extract a voice feature from an audio signal to generate a feature vector; a salience extractor configured to extract a importance of speech from at least one of the audio signal or a video signal to generate an importance vector; and a neural network configured to output a recognition result based on the feature vector and the importance vector, in which the recognition result is output by masking some.

Literature Review :

Title of Paper	Details of Publication with Date and Year	Literature Identified for Project
An Approach for Audio/Text Summary Generation from Webinars/Online Meetings.	Computational Intelligence and Communication Networks.2021	Text summary, audio summary, video to audio conversion. The proposed scheme is evaluated using short videos acquired from Youtube
AUDIO SUMMARIZATION FOR PODCASTS	29th European Signal Processing Conference (EUSIPCO). 2021	Converting podcast audio into test using AWS transcribe. A well-known ASR tool, PreSumm model.

Visual Audio Summarization Based on NLP Models	1st International Conference on Computational Science and Technology (ICCST). 2022	Utilizing Hugging-face's BART model for handling long range dependencies in sequence to sequence task. Presents deeper concept of NLP which is not required.
Text Summarization using NLP	International Research Journal of Engineering and Technology (IRJET). 2024	It covers extractive and abstractive summarization techniques, focusing on sentence selection and sentence scoring to retain key information.
Implementation of NLP based automatic text summarization using spacy	International Journal of Health Sciences.2022	The implementation ranks sentences by word frequency and importance, effectively summarizing the extracted text.
A survey on audio analysis: Text characterization and summarization -2024	World Journal of Advanced Research and Reviews	Comparative study of various algorithms such as Rule based systems, HMM, GANs, LSTM, RNN, BART model, Google speech to text api
Audio Transcription and Summarization System using Cloud Computing and Artificial Intelligence- 2023	International Journal on Recent and Innovation Trends in Computing and Communication	Evaluate various ASR services (Amazon Transcribe, Google Cloud Speech-to-Text, etc.) for accuracy, scalability, and features. Integration with ChatGPT: Uses GPT-3 to generate meeting summaries from transcribed text.
AUDIO DATA SUMMARIZATION SYSTEM USING NATURAL LANGUAGE PROCESSING- 2019	International Research Journal of Engineering and Technology (IRJET)	The system efficiently converts speech audio to text, summarizes it using NLP techniques like tokenization and sentence scoring, and provides both text and audio summaries of the content.
SPEECH RECOGNITION WITH DEEP RECURRENT NEURAL NETWORKS- 2013	IEEE International Conference on Acoustics, Speech and Signal Processing	Gives insights into how deep RNN are created for speech to text model and how various combination of layers affect the accuracy and errors
A Review on	5th International	Presents a workflow for

Sentiment Analysis of Text, Image and Audio Data - 2021	Conference on Computing Methodologies and Communication (ICCMC)	classification of audio data. Also, compare multiple models for better selection according to need.
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Current Limitations

1. The processing time is limited by the GPUs
2. It can only process English language with high accuracy. For other languages, accuracy would significantly affected due to non-availability of proper dataset and hardware.
3. Processing needs to be performed on the cloud due to hardware limitation on local desktops.
4. Each speaker needs to speak for atleast 5secs for diarization to succeed.
5. For summarization, theres needs to be minimum of 10 words spoken by speakers.

Proposed Solution

- 1.. Better hardware (GPU) support in future for local processing
- 2.. Generation and Utilization of proper language wise dataset and training these models on that generated data

Objectives and Scope of Work

Objectives

- Memory recall and Optimized task management
- Provide concise text summaries of their Daily interactions and meetings with given timestamps.

Scope of Work:

- For professionals, the system could summarize important points from meetings or conversations, helping users keep track of key decisions, action items, and follow-ups.
- Individuals could use the system as an audio journal, allowing them to reflect on their day's events. At the end of the day, they receive a concise summary of key conversations, thoughts, or important points captured.
- The system could aid in settings where a day's worth of audio recordings, such as in interviews, need to be summarized and stored as documentation for future reference.

Feasibility Assessment :

I. Expected Outcomes of the Project

1. Audio recording in .wav format on demand as needed by user using the switch to start and stop the audio recording instrument's switch
2. Processing the audio recorded to a text format where user is able to read what they had performed in their complete day
3. Giving user a summary of their daily activities as recorded by the audio recording instrument
4. Improve productivity of user by reducing time it takes to recall all the events in the day

II. Innovation Potential

The end product is cost effective due its reliance on easy to find hardware and OSS (Open Source Software). The Models used are always updated due to its open source capabilities.

III. Task Involved

1: Audio Data Preprocessing

- Collect and preprocess the dataset.
- Classification of speakers using speech diarization.
- Segmentation of the data.

2: Speech to Text (STT)

- Implement STT model for the dataset.
- Test the accuracy using WER(word error rate), CER(character error rate).
- Add post processing components such as annotations.

3: Text Preparation for Summarization

- Prepare and tokenize the text data.
- Remove punctuation.
- Perform necessary data cleaning and formatting.

4: Text Summarization

- Compile the cleaned data using RNN/transformer layers.
- Test and validate the summarization results.

5: Hardware Implementation

- A device consisting of microphone and memory card for recording.

IV. Expertise Required

1. Inhouse Expertise
 - Speech Recognition
 - Speaker diarization and classification
2. External Expertise
 - Voice Data Expert
 - NLP Researcher (if advanced summarization)
 - Cloud speech to text API consultant

V. Facilities Required

1. Inhouse Facilities
 - Performant Computing Setup
 - Python Development Enviroment
 - Speech to text libraries
 - Basic audio recording setup
2. External Facilities
 - Cloud GPU Services
 - Speech-to-Text APIs
 - External Hosting/Deployment Platforms

Milestones and Time Plan

	Task	JUL Y 2024	AU G 2024	SEP 2024	OC T 2024	NO V 2024	DE C 2024	JAN 2025	FEB 2025	MA R 2025	AP R 2025
Desi gn	Conceptual Design										
	Detailed design										
	Design Modificati ons										
	Final Design										
Dev elop	Procureme nt (If any)										
	Prototypin g										
	Modificati ons										
Deli ver	Testing and Validation										
	Final Modificati ons										
	IPR / patent draft										
	Thesis and Poster										

Name and Signature of Project Guide

Signature of HOD

SOCIAL IMPACT OF PROJECT

Developing and implementing more sophisticated social speech processing systems such as transcription, speaker diarization and summarization have considerable social ramifications that impact society much more than technological advancement itself. These systems make a great impact on social inclusion, accessibility, and diversity in the dissemination of information to numerous people and communities.

Perhaps the most profound social change impacts people with hearing disabilities and other communicational disabilities. Making it possible for speech to be converted into text in real-time makes it possible for anyone with physical disabilities to participate in conversations, lectures, meetings, and discussions at work and in public. Social discrimination by providing overwhelming communication blockades and the impossibility to access information and opportunities is eradicated.

Such system in education serve both students and teachers by providing real time accurate transcription of lectures and group discussions. Students coping with learning disabilities, second language learners and students from low resource classrooms benefit tremendously from these. Allowing the content to be revisited enriches the learners absorption of the material thus enhancing retention and academically.

The system ensures improved service in the business domain and organizational productivity by maintaining a record of every detail from meetings, brainstorming activities, and client meetings. It also accommodates remote and hybrid work models by providing clear, documented communication which aids collaboration among teams in different locations. In addition, it promotes organizational and transparency accountability with accurate records and summaries of meetings.

While the system may be beneficial in corporate settings, it has potential use in the legal and public service areas. Reliable transcription and speaker recognition can improve efficiency in courts, interviews, and public consultations. Having accurate documentation safeguards justice, governing, and civic participation.

Furthermore, the system enhances personal productivity by providing real-time audio summarization which reduces unnecessary information and time expenditures. This allows users to focus on more important activities rather than the grueling task of sifting through raw audio data.